

- ing under Dependency. *The Annals of Statistics*, 29(4):1165–1188.
- Uri Berger and Edoardo M. Ponti. 2024. [Cross-lingual and cross-cultural variation in image descriptions](#). *Preprint*, arXiv:2409.16646.
- Lucas Beyer, Andreas Steiner, André Susano Pinto, Alexander Kolesnikov, Xiao Wang, Daniel Salz, Maxim Neumann, Ibrahim Alabdulmohsin, Michael Tschannen, Emanuele Bugliarello, Thomas Unterthiner, Daniel Keysers, Skanda Koppula, Fangyu Liu, Adam Grycner, Alexey Gritsenko, Neil Houlsby, Manoj Kumar, Keran Rong, Julian Eisenschlos, Rishabh Kabra, Matthias Bauer, Matko Bošnjak, Xi Chen, Matthias Minderer, Paul Voigtlaender, Ioana Bica, Ivana Balazevic, Joan Puigcerver, Pinelopi Papalampidi, Olivier Henaff, Xi Xiong, Radu Soricut, Jeremiah Harmsen, and Xiaohua Zhai. 2024. [PaliGemma: A versatile 3B VLM for transfer](#). *Preprint*, arXiv:2407.07726.
- Helen Bird, David Howard, and Sue Franklin. 2003. [Verbs and nouns: The importance of being imageable](#). *Journal of Neurolinguistics*, 16(2):113–149.
- Walter Bisang. 2017. [Grammaticalization](#). In *Oxford Research Encyclopedia of Linguistics*. Oxford University Press.
- Geert Booij. 2007. [Inflection](#). In Geert Booij, editor, *The Grammar of Words: An Introduction to Linguistic Morphology*, pages 99–124. Oxford University Press.
- Marc Brysbaert, Amy Beth Warriner, and Victor Kuperman. 2014. [Concreteness ratings for 40 thousand generally known English word lemmas](#). *Behavior Research Methods*, 46(3):904–911.
- Xi Chen, Xiao Wang, Soravit Changpinyo, A. J. Piergiovanni, Piotr Padlewski, Daniel Salz, Sebastian Goodman, Adam Grycner, Basil Mustafa, Lucas Beyer, Alexander Kolesnikov, Joan Puigcerver, Nan Ding, Keran Rong, Hassan Akbari, Gaurav Mishra, Linting Xue, Ashish V. Thapliyal, James Bradbury, and Weicheng Kuo. 2023. [PaLI: A jointly-scaled multilingual language-image model](#). In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.
- Christine Chiarello, Connie Shears, and Kevin Lund. 1999. [Imageability and distributional typicality measures of nouns and verbs in contemporary English](#). *Behavior Research Methods, Instruments, & Computers*, 31(4):603–637.
- Norbert Corver and Henk Van Riemsdijk. 2001. [Semi-lexical categories](#). In Norbert Corver and Henk Van Riemsdijk, editors, *Semi-Lexical Categories*, pages 1–20. de Gruyter.
- Ryan Cotterell and Jason Eisner. 2017. [Probabilistic typology: Deep generative models of vowel inventories](#). In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1182–1192, Vancouver, Canada. Association for Computational Linguistics.
- Ryan Cotterell, Sabrina J. Mielke, Jason Eisner, and Brian Roark. 2018. [Are all languages equally hard to language-model?](#) In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 536–541, New Orleans, Louisiana. Association for Computational Linguistics.
- William Croft. 2002. *Typology and Universals*, 2nd edition. Cambridge University Press.
- Marie-Catherine de Marneffe, Christopher D. Manning, Joakim Nivre, and Daniel Zeman. 2021. [Universal Dependencies](#). *Computational Linguistics*, 47(2):255–308.
- Catherine Dubé, Laura Monetta, María Macarena Martínez-Cuitiño, and Maximiliano A. Wilson. 2014. [Independent effects of imageability and grammatical class in synonym judgement in aphasia](#). *Psicothema*, 26(4):449–456.
- Francis Ferraro, Nasrin Mostafazadeh, Ting-Hao Huang, Lucy Vanderwende, Jacob Devlin, Michel Galley, and Margaret Mitchell. 2015. [A survey of current datasets for vision and language research](#). In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 207–213, Lisbon, Portugal. Association for Computational Linguistics.
- Simeon Floyd. 2011. [Re-discovering the Quechua adjective](#). *Linguistic Typology*, 15(1):25–63.
- Spandana Gella, Rico Sennrich, Frank Keller, and Mirella Lapata. 2017. [Image pivoting for learning multilingual multimodal representations](#). In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 2839–2845, Copenhagen, Denmark. Association for Computational Linguistics.
- Team Gemma. 2024. [Gemma: Open models based on Gemini research and technology](#). *Preprint*, arXiv:2403.08295.
- Talmy Givon. 1984. *Syntax: A Functional-Typological Introduction Vol I*. Amsterdam: Benjamins.
- Joseph Harold Greenberg, editor. 1966. *Universals of Language*, 2nd edition. Number 37 in The M.I.T. Press Paperback Series. M.I.T Pr, Cambridge, Mass.
- John Hale. 2001. [A probabilistic Earley parser as a psycholinguistic model](#). In *Second Meeting of the North American Chapter of the Association for Computational Linguistics*.
- Coleman Haley, Edoardo M. Ponti, and Sharon Goldwater. 2024. [Corpus-based measures discriminate inflection and derivation cross-linguistically](#). *Journal of Language Modelling*, 12(2):477–529.

- Martin Haspelmath. 2007. [Pre-established categories don't exist: Consequences for language description and typology](#). *Linguistic Typology*, 11(1):119–132.
- Martin Haspelmath. 2010. [Comparative concepts and descriptive categories in crosslinguistic studies](#). *Language*, 86(3):663–687.
- Martin Haspelmath. 2012. [How to compare major word-classes across the world's languages](#). *UCLA Working Papers in Linguistics*, 17:109–130.
- Henrison Hsieh. 2019. [Distinguishing nouns and verbs: A Tagalog case study](#). *Natural Language & Linguistic Theory*, 37(2):523–569.
- Daniel Kaufman. 2009. [Austronesian Nominalism and its consequences: A Tagalog case study](#). *Theoretical Linguistics*, 35(1):1–49.
- Karin Kipper Schuler, Anna Korhonen, and Susan Brown. 2009. [VerbNet overview, extensions, mappings and applications](#). In *Proceedings of Human Language Technologies: The 2009 Annual Conference of the North American Chapter of the Association for Computational Linguistics, Companion Volume: Tutorial Abstracts*, pages 13–14, Boulder, Colorado. Association for Computational Linguistics.
- Roger Levy. 2008. [Expectation-based syntactic comprehension](#). *Cognition*, 106(3):1126–1177.
- Johan Liljencrants, Björn Lindblom, and Bjorn Lindblom. 1972. [Numerical Simulation of Vowel Quality Systems: The Role of Perceptual Contrast](#). *Language*, 48(4):839–862.
- Kimberly R Lin, Lisa Wisman Weil, Audrey Thurm, Catherine Lord, and Rhiannon J Luyster. 2022. [Word imageability is associated with expressive vocabulary in children with autism spectrum disorder](#). *Autism & Developmental Language Impairments*, 7.
- Fangyu Liu, Emanuele Bugliarello, Edoardo Maria Ponti, Siva Reddy, Nigel Collier, and Desmond Elliott. 2021. [Visually grounded reasoning across languages and cultures](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 10467–10485, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Dermot Lynott, Louise Connell, Marc Brysbaert, James Brand, and James Carney. 2020. [The Lancaster Sensorimotor Norms: Multidimensional measures of perceptual and action strength for 40,000 English words](#). *Behavior Research Methods*, 52(3):1271–1291.
- Joan Maling and So-Won Kim. 1998. Case assignment in the *sipta*-construction. In Ross King, editor, *Description and Explanation in Korean Linguistics*. East Asia Program, Cornell University, Ithaca, NY.
- Alireza Mohammadshahi, Rémi Lebret, and Karl Aberer. 2019. [Aligning multilingual word embeddings for cross-modal retrieval task](#). In *Proceedings of the Beyond Vision and LANGUAGE: inTEgrating Real-world kNowledge (LANTERN)*, pages 11–17, Hong Kong, China. Association for Computational Linguistics.
- Byung-Doh Oh and William Schuler. 2024. [Leading whitespaces of language models' subword vocabulary poses a confound for calculating word probabilities](#). *Preprint*, arXiv:2406.10851.
- Hyunji Hayley Park, Katherine J. Zhang, Coleman Haley, Kenneth Steimel, Han Liu, and Lane Schwartz. 2021. [Morphology matters: A multilingual language modeling analysis](#). *Transactions of the Association for Computational Linguistics*, 9:261–276.
- Andrew K. Pawley. 2006. Where have all the verbs gone? Remarks on the organisation of languages with small, closed verb classes. In *11th Biennial Rice University Linguistics Symposium*.
- Tiago Pimentel and Clara Meister. 2024. [How to compute the probability of a word](#). *Preprint*, arXiv:2406.14561.
- Tiago Pimentel, Clara Meister, Ethan Wilcox, Kyle Mahowald, and Ryan Cotterell. 2023. [Revisiting the optimality of word lengths](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2240–2255, Singapore. Association for Computational Linguistics.
- Frans Plank. 1994. Inflection and derivation. In *The Encyclopedia of Language and Linguistics*, pages 1671–1679. Elsevier Science and Technology, Amsterdam.
- Peng Qi, Yuhao Zhang, Yuhui Zhang, Jason Bolton, and Christopher D. Manning. 2020. [Stanza: A Python Natural Language Processing Toolkit for Many Human Languages](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 101–108, Online. Association for Computational Linguistics.
- Janarthanan Rajendran, Mitesh M. Khapra, Sarath Chandar, and Balaraman Ravindran. 2016. [Bridge correlational neural networks for multilingual multimodal representation learning](#). In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 171–181, San Diego, California. Association for Computational Linguistics.
- Alexander Rauhut. 2023. [Quantitative Aspects of the Word Class Continuum in English](#). Ph.D. thesis, Freie Universität Berlin.
- Norvin Richards. 2009. [Nouns, verbs, and hidden structure in Tagalog](#). *Theoretical Linguistics*, 35(1):139–152.

- Eva Schultze-Berndt. 2000. *Simple and Complex Verbs in Jaminjung: A Study of Event Categorisation in an Australian Language*. Ph.D. thesis, Radboud University, Nijmegen.
- Graham G. Scott, Anne Keitel, Marc Becirspahic, Bo Yao, and Sara C. Sereno. 2019. [The Glasgow Norms: Ratings of 5,500 words on nine scales](#). *Behavior Research Methods*, 51(3):1258–1270.
- Nathaniel J. Smith and Roger Levy. 2013. [The effect of word predictability on reading time is logarithmic](#). *Cognition*, 128(3):302–319.
- Adrian Staub. Forthcoming. [Predictability in Language Comprehension: Prospects and Problems for Surprisal](#). *Annual Review of Linguistics*.
- Ashish V. Thapliyal, Jordi Pont Tuset, Xi Chen, and Radu Soricut. 2022. [Crossmodal-3600: A Massively Multilingual Multimodal Evaluation Dataset](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 715–729, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Henri Theil. 1970. [On the Estimation of Relationships Involving Qualitative Variables](#). *American Journal of Sociology*, 76(1):103–154.
- David John Weber. 1983. *A Grammar of Huallaga (Huanuco) Quechua*. Ph.D. thesis, University of California, Los Angeles, United States – California.
- Ethan G. Wilcox, Tiago Pimentel, Clara Meister, Ryan Cotterell, and Roger P. Levy. 2023. [Testing the predictions of surprisal theory in 11 languages](#). *Transactions of the Association for Computational Linguistics*, 11:1451–1470.
- Tianxing Wu, Chaoyu Gao, Lin Li, and Yuxiang Wang. 2022. [Leveraging multi-modal information for cross-lingual entity matching across knowledge graphs](#). *Applied Sciences*, 12(19).
- Andre Ye, Sebastin Santy, Jena D. Hwang, Amy X. Zhang, and Ranjay Krishna. 2024. [Computer vision datasets and models exhibit cultural and linguistic diversity in perception](#). *Preprint*, arXiv:2310.14356.
- X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer. 2023. [Sigmoid loss for language image pre-training](#). In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11941–11952, Los Alamitos, CA, USA. IEEE Computer Society.